PAGE NO.:

STATISTICS

- Measures of central tendancy

→ Arithmetic mean
→ Median
→ Mode
→ Geometric mean
→ Harmonic mean

- Measures of Dispersion

→ Range : UB - LB
→ Quartile deviation → Q_1, Q_2, Q_3, Q_4 : $\frac{Q_3 - Q_1}{2}$
→ Decile : $D_1, D_2, \ldots, D_{10}$
→ Standard deviation (σ^2/N) $\sigma = \sum (x_i - \bar{x})^2$
→ Mean deviation → about mean, median, mode

- Measure of skewness
- Measure of Kurtosis : Measures thickness

Inclusive class interval

marks	f
8-9	1
10-11	3
12-13	1
14-15	2

Exclusive class interval

marks	f
7.5-9.5	1
9.5-11.5	3
11.5-13.5	1
13.5-15.5	2

- Data
→ It can be defined as a systematic record of a particular quantity. • Depending on the nature of the source of data it is further categorised as primary & secondary data.

1) Primary data: Data that is collected for the first time by an investigator for a specific purpose. Primary data is pure in the sense that no statistical operations have been performed on them and they are original.
eg. census of India

2) Secondary data: Data that is sourced from some places that have originally collected it, i.e the data is already collected by some researchers in the past and is available either in published or unpublished form. This type of data is impure as the statistical operations may have been performed already on them.
eg. Finance website of government of India.

- Depending on the nature of the observation:
a) Qualitative data and b) Quantitative data

a) Qualitative data: It represents some characteristics or attributes, they depict descriptions that may be observed but cannot be ~~calle~~ calculated.
eg. Honesty, Wisdom, Intelligence

b) Quantitative data: It can be measured and not simply observed. It can be numerically represented and calculations can be performed on them.
eg: marks, height

• Depending on the ~~nature~~ representation of the observation:

a) Discrete data:
b) continuous data

a) Discrete data: Data than can take only certain specific values rather than a range of the values.
Eg: data of blood group of certain population.

b) Continuous data: Data that can values between a certain range with the highest and lowest values.
The difference between the upper limit and lower limit is called the class interval of that class and the difference between ~~t~~ highest and lowest values is called range of the data.

Note: 1] Discrete data is usually ~~#~~ represented by using bar charts.
2] Continuous data can be represented using histographs.

Measures of central tendancy or average

It gives us an idea about the concentration of the values in the central part of the distribution.
It is the value of the variable which is representative of the entire value of distribution
The following are the five measures of central tendancys
1) Arithmatic mean
2) Median
3) Mode
4) Geometric mean
5) Harmonic mean.

Basics assumptions of an ideal measure of central tendancy

⇒ 1) It should be rigidly defined.

2) Ready comprehensible and easy to calculate

3) It should based all observations

4) It should be suitable for further mathenitical treatment.

5) It should be affected as little as possible by fluctuations of sampling.

6) It should not be affected much by extreme values

- Arithmetic mean

$x_1, x_2, x_3 \dots x_n$ are observations

$$\bar{x} = \frac{\sum x_i}{n}, \quad n = \text{no. of observations} \rightarrow \text{ungrouped}$$

or

$$\bar{x} = \frac{\sum x_i f_i}{\sum f_i} \rightarrow \text{grouped}$$

Q) Find mean of marks obtained

Roll no.	01	02	03	04	05
marks	20	22	25	28	30

$$N = 5, \quad \bar{x} = \frac{\sum x_i}{N} = \frac{20+22+25+28+30}{5}$$

$$\bar{x} = 25$$

Q] Find average marks using mean

marks	8	10	15	20
freq.	5	8	8	4

x_i	f_i	$x_i f_i$
8	10	80
10	8	80
15	8	120
20	4	80
	$\sum f_i = 25$	$\sum x_i f_i = 320$

$$\bar{x} = \frac{\sum x_i f_i}{\sum f_i} = \frac{320}{25} = 12.8 \text{ marks}$$

Q] Find mean marks

marks	no. of stud. (f_i)	x_i	$x_i f_i$
0-10	7	5	35
10-20	8	15	120
20-30	20	25	500
30-40	10	35	350
40-50	5	45	225
Total	50		1230

$$\bar{x} = \frac{\sum x_i f_i}{\sum f_i} = 24.6$$

* also, $\bar{x} = \underbrace{A + \frac{\sum f_i d_i}{\sum f_i}}_{\text{Assumed mean}}$, where, $d_i = x_i - A$

eg→ 1)

x_i	f_i	$d_i = x_i - 10$	$f_i d_i$
8	10	-2	-10
10	8	0	0
15	8	5	40
20	4	10	40
	$\Sigma f_i = 25$		$\Sigma f_i d_i = 70$

→ Change of origin

$$\bar{x} = A + \frac{\Sigma d_i f_i}{\Sigma f_i} = 10 + \frac{70}{25} = 12.8$$

2)

x_i	$d_i = x_i/1000$
1000	1
2000	2
3000	3
4000	4
5000	5

→ Change of scale

$$\frac{\Sigma d_i}{5} = \bar{d} = 5$$

$\therefore \quad \bar{x} = h \times \bar{d} = 1000$

$h = 1000$

$\bar{x} = 2000$

If, $d_i = \dfrac{x_i - A}{h}$

$$\bar{x} = A + \left(h \times \frac{\Sigma d_i f_i}{\Sigma f_i} \right)$$

* Calculate the mean for the following using assumed mean

Class interval	0-8	8-16	16-24	24-32	32-40	40-48
frequency	8	7	16	24	15	7

3)

C.I	f_i	x_i	$x_i - A$	$d_i = \frac{x_i - A}{h}$	$d_i f_i$
0-8	8	4	-16	-2	-16
8-16	7	12	8	-1	-7
16-24	16	20	0	0	0
24-32	24	28	8	1	24
32-40	15	36	16	2	30
40-48	7	44	24	3	21
	$\Sigma f_i = 77$				$\Sigma f_i d_i = 52$

$A = 20$ and $h = 8$

$\therefore \bar{x} = A + \left(h \times \frac{\Sigma d_i f_i}{\Sigma f_i}\right) = 20 + 8 \times \frac{52}{77} = 25.4$

4)

x_i	f_i	$x_i - A$	$d_i = \frac{x_i - A}{h}$	$d_i f_i$
50	3	-150	-3	-9
100	4	-100	-2	-8
150	9	-50	-1	-9
200	1	0	0	0
250	2	50	1	2
300	1	100	2	2
	20			-22

$A = 200, \; h = 50$

$\bar{x} = 200 + 50 \times \left(\frac{-22}{20}\right) = 200 - 55 = 145$

discrete grouped me median ke liye $\frac{N}{2}$ nikalke uske just bad wala.

PAGE NO.:

Median

→ denote it by Me

1) arrange in ascending and descending
2) If N is odd ⇒ $Me = \left(\frac{N+1}{2}\right)^{th}$ term

If N is even ⇒ Me = any value between $\left(\frac{N}{2}\right)^{th}$ term & $\left(\frac{N}{2}+1\right)^{th}$ term

Q) Find median of 10, 11, 12, 13, 6, 8, 9, 10, 13

6, 8, 9, 10, 10, 11, 12, 13, 13

N = 9, $Me = \left(\frac{9+1}{2}\right)^{th}$ term $= 5^{th}$ term

Median = 10

Q) Find median of 10, 11, 12, 13, 6, 8, 9, 10

6, 8, 9, 10, 10, 11, 12, 13

N = 8, $Me = \left(\frac{8}{2}\right)^{th}$ term $= 4$, $\left(\frac{8}{2}+1\right)^{th}$ term $= 5$

∴ median = 10

Q)

x_i	f_i	c.f
6-8	1	1
8-10	3	4
10-12	3	7
12-14	4	11

Now, $Me = 10 + \frac{2}{3}\left(\frac{11}{2} - 4\right) =$

$$\text{Median} = l + \frac{h}{f}\left(\frac{N}{2} - c\right)$$

l = lower limit of median class
h = class width of median class
f = frequency of median class
N = Total frequency
c = Cf of the class preceding the median class

Mode

In the data which frequency is greater that is called mode, but for the cases,
Case I: If the maximum freq. is repeated
Case II: If max. freq occurs in very beginning or at the end.
Case III: If there are irregularities in the distribution
Then the method of grouping is helpful to calculate mode.

Q]

x	f (i)	(ii)	(iii)	(iv)	(v)	(vi)
1	3	11		26		
2	8		23		46	
3	15	38				73
4	23		58	98		
5	35	75			107	
6	40		72			100
7	32	60		80		
8	28		48		51	
9	20	65				79
10	45		59	65		
11	14	20				
12	6					

PAGE NO.:

i	ii	iii	iv	v	vi
45	75	72	98	107	100
10	5,6	6,7	4,5,6	5,6,7	6,7,8

4	5	6	7	8	10
1	111	𝍸	111	1	1

∴ mode is 6

For grouped continuous data
First identify modal class

$$M_o = l + \frac{h\,(f_1 - f_0)}{(f_1 - f_0) - (f_2 - f_1)}$$

• Find mean by step deviation, mode and median.

marks	fi	ii	iii	iv	v	vi	xi	cf	di	difi
0-10	3	11					5	3	-5	-15
10-20	8		23	26			15	10	-4	-32
20-30	15	38			46		25	26	-3	-45
30-40	23		58			73	35	49	-2	-46
40-50	35	75		98			45	84	-1	-35
50-60	40		72		107		55	124	0	0
60-70	32	60				100	65	156	1	32
70-80	28		48	80			75	184	2	56
80-90	20	65			451		85	204	3	60
90-100	45		59			79	95	249	4	180
100-110	14	20		65			105	253	5	70
110-120	6						115	269	6	36
	269									261

Mode →

	i	ii	iii	iv	v	vi
max freq	45	75	72	98	107	100
marks	C_{10}	C_5, C_6	C_6, C_7	C_4, C_5, C_6	C_5, C_6, C_7	C_6, C_7, C_8

modal class $= C_6$

$C_6 \rightarrow 50-60$

$\therefore$ mode $= l + \frac{h(f_1 - f_0)}{(f_1-f_0)-(f_2-f_1)}$

$= 50 + \frac{50}{13}$

$= 50 + 3.846$

$= 53.846$

Median $\rightarrow \frac{N}{2} = \frac{269}{2} = 134.5$

$C_7 \mapsto$ median class

median $= l + \frac{h}{f}\left(\frac{N}{2} - C\right)$

$= 60 + \frac{10}{32}(134.5 - 124)$

$= 60 + \frac{10}{32} \times 10.5$

$= 60 + \frac{5}{16} \times 10.5$

$= 63.281$

Mean $\rightarrow$ $A + h\left(\frac{\Sigma d_i f_i}{\Sigma f_i}\right)$

$A = 55$

$h = 10$

$\therefore$

mean $= 55 + 10\left(\frac{261}{269}\right)$

≈ 64.70

Partition

→ These are the values which divide the series into a number of equal parts

1) Quartiles: The three points which divide the series into four equal parts are called quartiles. They are denoted as Q_1, Q_2, Q_3 respectively. The first quartile Q_1 is the value which exceed 25% of the observation and is exceeded by 75% of the observation. The second quartile Q_2 coincides with the median. The third quartile Q_3 is the point which has 75% observations before it and 25% observations after it. Q_1, Q_2, Q_3

2) Decile: The nine points which divide the series into ten equal parts are called deciles. First decile is denoted as D_1, second as D_2, D_5 represents / coincides median. D_1, D_2, D_3 . . . D_9,

3) Percentile: The ninety-nine points which divide the series into hundred equal parts are called percentiles. It is denoted as P_1, P_2, P_3 . . . . P_{99}. For example, P_{47} is the 47^{th} percentile, is the point which exceeds 47% of the observation

The 7^{th} decile D_7 has 70% observations before it

$$\boxed{Q_2 = D_5 = P_{50} = \text{median}}$$

The method of computation of the partition values is the same as that of the median.

- Quartile 1:

compare $\frac{N}{4} \leq C$, gives Q_1 class

$$Q_1 = l + \frac{h}{f}\left(\frac{N}{4} - C\right)$$ [c is preceding c.f to the Q_1 class]

- Quartile 3:

compare $\frac{3N}{4} \leq C$, gives Q_3 class

$$Q_3 = l + \frac{h}{f}\left(\frac{3N}{4} - C\right)$$ [C is preceding cf to the Q_3 class]

- Decile 1:

compare $\frac{N}{10} \leq C$, gives D_1 class

$$D_1 = l + \frac{h}{f}\left(\frac{N}{10} - C\right)$$ [c is preceding cf to the D_1 class

⋮

- Decile 10:

compare $\frac{9N}{10} \leq C$, gives D_{10} class

$$D_{10} = l + \frac{h}{f}\left(\frac{9N}{10} - C\right)$$ [c is the preceding cf to the D_{10} class]

⋮

- Percentile 3:

compare $\frac{N}{33} \leq C$, gives P_3 class

$$P_3 = l + \frac{h}{f}\left(\frac{N}{33} - C\right)$$ [c is the preceding cf to the P_3 class]

PAGE NO.:

- Quartile 1:

  compare $\frac{N}{4} \leq C$, gives Q_1 class

  $$Q_1 = l_1 + \frac{h}{f}\left(\frac{N}{4} - C\right)$$ [c is preceding c.f to the Q_1 class]

- Quartile 3:

  compare $\frac{3N}{4} \leq C$, gives Q_3 class

  $$Q_3 = l + \frac{h}{f}\left(\frac{3N}{4} - C\right)$$ [C is preceding cf to the Q_3 class]

- Decile 1:

  compare $\frac{N}{10} \leq C$, gives D_1 class

  $$D_1 = l + \frac{h}{f}\left(\frac{N}{10} - C\right)$$ [c is preceding cf to the D_1 class

  ⋮

- Decile 10:

  compare $\frac{9N}{10} \leq C$, gives D_{10} class

  $$D_{10} = l + \frac{h}{f}\left(\frac{9N}{10} - C\right)$$ [c is the preceding cf to the D_{10} class]

  ⋮

- Percentile 3:

  compare $\frac{N}{33} \leq C$, gives P_3 class

  $$P_3 = l + \frac{h}{f}\left(\frac{N}{33} - C\right)$$ [c is the preceding cf to the P_3 class]

x	f	cf
0	1	1
1	9	10
2	26	36
3	59	95
4	72	167
5	52	219
6	29	248
7	7	255
8	1	256

Find, $Q_1, Q_2, Q_3, D_1, D_2, D_3 \ldots D_{10}, P_{30}, P_{45}, P_{75}, P_{90}, P_{99}$

1) Q_3,

$\frac{3N}{4} = 192$

$\therefore\ C = 219$

$\therefore\ Q_3 = 5$

2) Q_1

$\frac{N}{4} = 64$

$\therefore\ C = 95$

$Q_1 = 3$

3) D_3

$\frac{3N}{10} = 76.8$

$C = 95$

$D_3 = 3$

4) P_{30}

$\frac{30N}{100} = 76.8$

$C = 95$

$P_{30} = 3$

5) P_{75}

$\frac{75N}{100} = 192$

$C = 219$

$P_{75} = 5$

6) P_{90}

$\frac{90N}{100} = 230.4$

$C = 248$

$P_{90} = 7$

7) Q_2

$\frac{N}{2} = 128$

$Q_1 = 4$

8) P_{99}

$\frac{99N}{100} = 253.44$

$P_{99} = 8$

9) P_{45}

$\frac{45N}{100} = 115.2$

$P_{45} = 4$

Measures of Dispersion

→ It gives an idea about homogenity or heterogenity of the distribution. Following are the types of measures of dispersion,

1) Range
2) Quartile deviation.
3) Mean deviation
4) Standard deviation & Root mean square deviation.

• Characteristics of an ideal measure of dispersion

1) It should be rigidly defined.
2) Easy to calculate and understand.
3) Should be based on all the observations.
4) Should be utilized to further mathematical treatment.
5) It should be affected as easy as possible by fluctuation of sampling.

* Range

→ Range is the difference between two extreme observations of the distribution.

$$Range = X_{max} - X_{min}$$

* Quartile deviation

→ The another name is semi-interquartile range. It is denoted and calculated by,

$$Q = \frac{1}{2}(Q_3 - Q_1)$$, Q_1 & Q_3 are 1st & 3rd quartiles

Note that it is the better measure than range as it makes use of 50% of data.

★ Mean deviation

→ If $x_i | f_i$ is the distribution then mean deviation about the average value A (mean/ median/ mode) is given by :-

$$\frac{\Sigma_i f_i |x_i - A|}{\Sigma_i f_i}$$

a) When $A = \bar{x}$, then it is called mean deviation about mean.

b) when $A = Me$, then it is called mean deviation about median.

c) When $A = Mo$, then it is called mean deviation about mode.

★ Standard deviation

→ It is usually denoted as 'σ', it is the positive square root of the arithmetic mean of the squares of the deviations of the given values from their arithmetic mean.

$$\sigma = \sqrt{\frac{\Sigma f_i (x_i - \bar{x})^2}{\Sigma f_i}} = \sqrt{\frac{1}{N}\left(\Sigma f_i (x_i - \bar{x})^2\right)}$$

Variance

The square of standard deviation is variance (σ^2)

★ Root mean square deviation

$$s = \sqrt{\frac{\Sigma f_i (x_i - A)^2}{\Sigma f_i}}$$

Jab $A = \bar{x}$ matlab A about mean raha to standard deviation aur RMS deviation equal.

PAGE NO.:

* Mean square deviation (s^2)
→ square of RMS deviation.

* Different formulae for calculating variance (σ^2) is:

$$\sigma^2 = \frac{1}{N}\left(\sum f_i x_i^2\right) - \left(\frac{1}{N}\sum f_i x_i\right)^2$$

or

$$\sigma^2 = \left(\frac{1}{N}\sum f_i x_i^2\right) - (\bar{x})^2$$

$$\sigma^2 = \frac{1}{N}\sum f_i (x_i - \bar{x})^2$$

Prove it

* Step deviation method for calculating σ^2

Variance is independent of change of origin but not of scale.
If $d_i = \frac{x_i - A}{h}$

then $\sigma_x^2 = h^2 \sigma_d^2$

σ_x = standard deviation of $x_i f_i$
σ_d = standard deviation of $d_i f_i$

* Variance of combined series,
If n_1, n_2 are the sizes of two distributions and $\bar{x}_1, \bar{x}_2$ are their means then combined series mean $\bar{x}$ is given by,

$$\bar{x} = \frac{n_1 x_1 + n_2 x_2}{n_1 + n_2}$$

and combined series variance is given by,

$$\sigma^2 = \frac{1}{n_1+n_2}\left[n_1(\sigma_1^2 + d_1^2) + n_2(\sigma_2^2 + d_2^2)\right]$$

σ_1, σ_2 are standard deviation of series 1 and 2.
$d_1 = \bar{x}_1 - \bar{x}$, $d_2 = \bar{x}_2 - \bar{x}$

Calculate mean and S.D using change of scale and change of scale, also calculate CD and CV

Ages	No. of members	x_i	$d_i = \frac{x_i - 55}{10}$	f_id_i	d_i^2	$f_id_i^2$
20-30	3	25	-3	-9	9	27
30-40	61	35	-2	-122	4	244
40-50	132	45	-1	-132	1	132
50-60	153	55	0	0	0	0
60-70	140	65	1	140	1	140
70-80	51	75	2	102	4	204
80-90	2	85	3	6	9	18
				-15		765

mean: $\bar{x} = A + h \frac{\Sigma f_i d_i}{\Sigma f_i}$

$= 55 + \frac{10(-15)}{542}$

$= 54.72$ years

S.D – $\sigma_{x}^2 = h^2 (\sigma_d^2)$

$\sigma_d^2 = \frac{1}{N} \Sigma f_i d_i^2 - \left(\frac{1}{N} \Sigma f_i d_i\right)^2$

$= \frac{765}{542} - (0.027)^2$

$= 1.41143911 44 - 0.000729$

$= 1.4107$

PAGE NO.:

$\sigma_x^2 = h^2 \sigma_d^2$

$\sigma_x^2 = (100)(1.41)$

$\sigma_x^2 = 141.07$

$\sigma_x = 11.877 = 11.88$ years

$CV = 100 \times \frac{11.88}{54.72}$

$= 21.7105$

$CD = \frac{\sigma}{\bar{x}} = 0.2171$

Coefficient of Dispersion

→ To compare the variability of two ~~units~~ series which are measured in different units. We do not calculate the measures of dispersion but we calculate coefficients of dispersion which are pure numbers independant of unit. Depending on measures of dispersion we have the following coefficients of dispersion:-

1) CD based on range →

$$\frac{X_{max} - X_{min}}{X_{max} + X_{min}}$$

2) CD based on quartile deviation:-

$$CD = \frac{(Q_3 - Q_1)/2}{(Q_3 + Q_1)/2}$$

3) CD based on mean deviation :-

$$CD = \frac{\text{mean deviation about A}}{\text{average (from which it is calculated)}}$$

4) CD based on standard deviation :-

$$CD = \frac{\sigma}{\bar{x}}$$

5) ~~CD~~ Coefficient of variation → $100 \times \frac{\sigma}{\bar{x}} = C.V$

- Moments about any point $x = A$:-

The r^{th} moment of a distribution about a point $x = A$ is denoted by $\mu_r'(x = A)$

$$\mu_r' = \frac{1}{N} \sum f_i (x_i - A)^r \quad , N = \sum f_i$$

First moment $\Rightarrow \mu_1' = \frac{1}{N} \sum f_i (x_i - A)$

Second moment $\Rightarrow \mu_2' = \frac{1}{N} \sum f_i (x_i - A)^2$

- Moments about mean

The r^{th} moment about mean is denoted by μ_r and it is the value of

$$\mu_r = \mu_r'(x = \bar{x}) = \frac{1}{N} \sum f_i (x_i - \bar{x})^r$$

$$\mu_1 = \frac{1}{N} \sum f_i (x_i - \bar{x})$$

$$\mu_r = \frac{1}{N} \sum f_i (x_i - \bar{x})^2 = \text{variance}$$

Q] Find the first four moments about $x = 55$, i.e. $\mu_1'(x=55)$ ...

Also find first four moments about mean μ_1, μ_2 ...

Ages	f_i	x_i	$x_i - 55$	$x_i - \bar{x}$	$f_i(x_i - A)$	$f_i(x_i - \bar{x})$
20-30	3	25	-30	-29.72	-90	-89.16
30-40	61	35	-20	-19.72	-1220	-1202.92
40-50	132	45	-10	-9.72	-1320	-1283.04
50-60	153	55	0	0	0	0
60-70	140	65	10	10.28	1400	1439.2
70-80	51	75	20	20.28	1020	1034.28
80-90	2	85	30	30.28	60	60.56

CA last ques

Q. For a group of 200 candidate, $\bar{x} = 40$ and $\sigma = 15$, later on it was discovered that the marks 43 and 35 were misread as 34 and 53 resp. Find corrected $\bar{x}$ and σ for corrected figures.

→ $\bar{x} = \frac{\Sigma x_i}{N} = \frac{\Sigma x_i}{200} = 40$

$\Sigma x_i = 8000$

corrected $\Sigma x_i = 8000 + (43+35) - 34 - 53 = 7991$ — (i)

$\therefore$ corrected $\bar{x} = \frac{\text{corrected } \Sigma x_i}{200}$

$= \frac{7991}{200} = 39.955$ score

$\sigma^2 = \frac{1}{N} \Sigma (x_i - \bar{x})^2$

$\sigma^2 = \frac{1}{N} \Sigma x_i^2 - \left(\frac{1}{N} \Sigma x_i\right)^2$

$\therefore$ for corrected $\sigma^2 = \frac{1}{N}(\text{corrected } \Sigma x_i^2) - \frac{1}{N}$ (corrected Σx_i

Now we calculate corrected Σx_i^2

$= (\text{incorrected } \Sigma x_i^2) - (34^2 + 53^2) + (43^2 + 35^2)$

$\therefore 15^2 = \frac{\Sigma x_i^2}{N} - (40)^2$

$\frac{\Sigma x_i^2}{N} = 1825$

incorrect $\Sigma x_i^2 = 365000$

$\therefore$ correct $\Sigma x_i^2 = 365000 - 1156 - 2809 + 1849 + 1225$

$= 364109$

PAGE NO.:

$\therefore \sigma^2 = \frac{364109}{200} - 1596.40$

$\sigma^2 = 1820.54 - 1596.40$

$\sigma^2 = 224.145$

$\sigma = 14.97$

#

- Relation between moments about mean (μ_r) and moments about any point ($x=A$), μ_r' ($x=A$)

$$\mu_2 = \mu_2' - (\mu_1')^2$$
$$\mu_3 = \mu_3' - 3\mu_2'\mu_1' + 2\mu_1'$$
$$\mu_4 = \mu_4' - 4\mu_3'\mu_1' + 6\mu_2'\mu_1'^2 - 3\mu_1^4$$

Measures of skewness.

Have an idea about the shape of the curve for the given distribution we study skewness.

A distribution is said to skewed if;

1) $\bar{x} \neq Me \neq Mo$
2) Quartiles are not equidistant from median.
$$Q_3 - Me \neq Me - Q_4$$
3) The curve is not symmetrical but is streched more to one side than other.

3 types:-

1) $S_k = \bar{x} - Me$
2) $S_k = \bar{x} - Mo$
3) $S_k = (Q_3 - Me) - (Me - Q_1)$

Coefficient of measure of skewness (C.S)

I) Karl Peason's coefficient

(CS) $S_k = \frac{\bar{x} - M_o}{\sigma}$

or $S_k = \frac{3(\bar{x} - Me)}{\sigma}$, as $Mo = 3Me - 2\bar{x}$

II) Bowley's coefficient

(CS) $S_k = \frac{(Q_3 - Me) - (Me - Q_1)}{(Q_3 - Me) + (Me - Q_1)}$

Karl Peason's β & γ coefficients

$\beta_1 = \frac{\mu_3^2}{\mu_2^3}$, $\gamma_1 = \sqrt{\beta_1}$

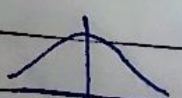
$\beta_2 = \frac{\mu_4}{\mu_2^2}$, $\gamma_2 = \beta_2 - 3$

III) Based upon moments CS

$S_k = \frac{\sqrt{\beta_1}\,(\beta_2 + 3)}{2(5\beta_2 - 6\beta_1 - 9)}$

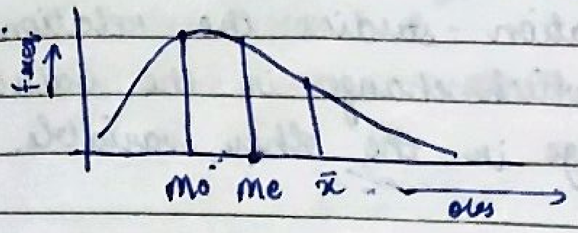
Symmetric distribution

If in a distribution $\bar{x} = Me = Mo$, then it is symmetric distribution

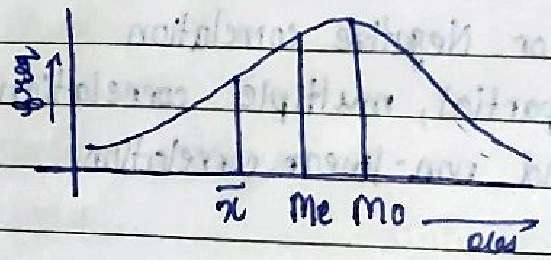


$Me = Mo = \bar{x}$

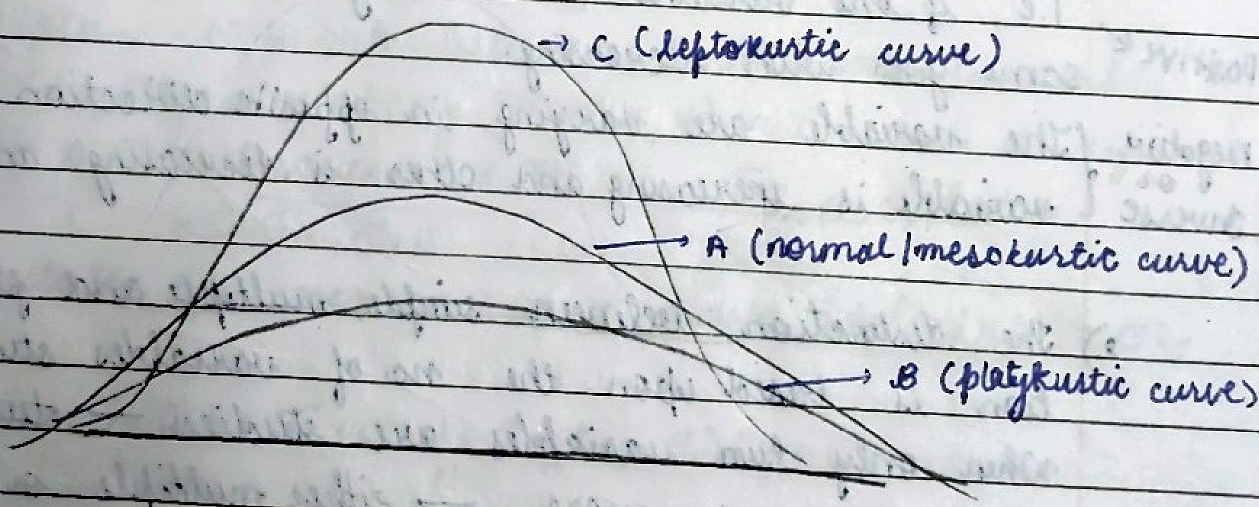
- Positively skewed distribution :
If in a distribution $Mo < Me < \bar{x}$, then it is positively skewed distribution.



- Negatively skewed distribution.
$Mo > Me > \bar{x}$



Measure of Kurtosis



C → $\beta_2 > 3$, $\gamma_2 > 0$
A → $\beta_2 = 3$, $\gamma_2 = 0$
B → $\beta_2 < 3$, $\gamma_2 < 0$

Corelation

~~Correlate~~

Corelation studies the relationship between two values in which change in the value of one variable cause change in the other variable.

→ Classification of corelation

1) Positive or Negative correlation
2) Simple, partial, multiple correlation
3) Linear and non-linear correlation

1) If both the variable are varying in the same direction i.e if one variable is increasing then other is also increasing same goes with decreasing. ← ...sitive

...egative or ...verse { The variable are varying in opposite direction i.e one variable is increasing and other is decreasing and vice versa

2) The distinction between simple, multiple and partial corelation is based upon the no. of variables studied. ~~Wt~~ When only two variables are studied → ~~studie~~ simple
three or more → either multiple or partial
In multiple correlation three or more variables are studied simultaneously on the other hand in partial correlation we recognize more than two variables, but consider only two variables to be influencing each other and the effect of other influencing variables being kept constant.

3) If the amount of change in one variable tends to bear constant ratio to the amount of change in other variable then correlation is said to be linear.
And opposite to this is non linear correlation/curvylinear.

★ Method of obtaining correlation coefficient.

1] Scatter digram method
2] Graphical method
3] Karl Pearson's correlation coefficient
4] Method of least square

• Karl Pearson's correlation coefficient

→ correlation coefficient between two random variable X and Y usually denoted by $r(X, Y)$ or r_{XY} and is defined by $r(X, Y) = \frac{\text{Covariance}(X, Y)}{\sigma_X \sigma_Y}$ or $\frac{\sigma_{XY}}{\sigma_X \sigma_Y}$, CoV is calculated as covariance of X and Y, i.e $\text{CoV}(X, Y) = \sigma_{XY}$

$$\text{CoV}(X, Y) = \sigma_{XY} = \frac{1}{N} \sum f_i (x_i - \bar{x})(y_i - \bar{y})$$

$$= \frac{1}{N} \sum f_i x_i y_i - \bar{x}\bar{y}$$

NOTE 1: The value of $-1 \leq r_{XY} \leq 1$
2: If $r_{XY} = +1$ then the correlation is said to be perfect and positive correlation.
3: If $r_{XY} = -1$, then perfect and negative, 0 → no correlation

Result: correlation coefficient is independent of change of origin and change of scale.

i.e If we have to find r_{xy} then we can conviniently shift/apply the change of origin and scale by:

$$U = \frac{x-A}{h}, \quad V = \frac{y-B}{k}$$

hence $r_{xy} = r_{uv}$

$$r_{uv} = \frac{Cov(U,V)}{\sigma_u \sigma_v} = \frac{\frac{1}{N}\Sigma f_i u_i v_i - \bar{u}\bar{v}}{\left(\frac{1}{N}\Sigma f_i u_i^2 - (\bar{u}^2)\right)^{1/2}\left(\frac{1}{N}\Sigma f_i v_i^2 - (\bar{v})^2\right)^{1/2}}$$

~~[illegible]~~

Q.1) calculate the correlation coefficient between following variable

= 687
= 68

x	y	$x_i y_i$	x_i^2	y_i^2	$x_i - A$	$x_i - B$	$u_i v_i$	u_i^2	v_i^2
65	67	4355	4225	4489	-2	-1	2	4	1
66	68	4488	4356	4624	-1	-0	0	1	0
67	65	4355	4489	4225	0	-3	0	0	9
67	68	4556	4489	4624	0	-0	0	0	0
68	62	4216	4624	3844	1	-6	-6	1	36
69	62	4278	4761	3844	2	-6	-12	4	36
70	69	4830	4900	4761	3	1	3	9	1
72	71	5112	5184	5041	5	3	15	25	9

$$\frac{Co(x,y)}{\sigma_x \sigma_y} = r_{xy}$$

$$\therefore Co(x,y) = \frac{1}{N}\Sigma x_i y_i - \bar{x}\bar{y}$$

$\bar{x} = 68$, $\bar{y} = 66.5$

$$Co(x,y) = \frac{36190}{8} - (68)(66.5)$$

$$= 4523.75 - 4522$$

$$= 1.75$$

$$\sigma x^2 = \frac{1}{N} \Sigma x_i^2 - (\bar{x})^2$$

$$= \frac{37028}{8} - 4624$$

$$= 4628.5 - 4624$$

$$\sigma x = \sqrt{4.5} = 2.121$$

$$\sigma y^2 = \frac{1}{N} \Sigma y_i^2 - (\bar{y})^2$$

$$= \frac{35452}{8} - 4422.25$$

$$= 4431.5 - 4422.25$$

$$\sigma y = \sqrt{9.25} = 3.041$$

$$\therefore \cancel{r_{xy} = \frac{1.75}{4.5 \times 9.25} = \frac{1.75}{41.625} = 0.042}$$

$$r_{xy} = \frac{1.75}{2.121 \times 3.041} = \frac{1.75}{6.449} = 0.271$$

• Rank correlation
Let us suppose that a group of n individuals is arranged in order of merit on the basis of two characteristics X and Y. These ranks in two characteristics in general will be different. Let us denote as (x_i, y_i). Then the Pearsonian coefficient of correlation ~~is~~ ~~called~~ b/w X and Y is called as the rank correlation coefficient.
We will denote it by $\rho = 1 - \left(\frac{\sum d_i^2 \times 6}{n(n^2-1)}\right)$, where $d_i = x_i - y_i$

Regression
Regression is the measure of the average relationship between two or more variables in terms of the original units of the data.
Types of variables in regressional analysis →
- The variable whose value is to be predicted is called dependant variable (regressed variable / explained variable).
- The variable which is used for prediction is called independant variable (regressor / predictor / explainatory variable)

$x = a + by$ (bxy)
$y = a + bx$ (byx)

PAGE NO.:

★ Linear and curvylinear Regression :-

→ If the variable in the bivariate distribution are related then we will find that the points in the scatter diagram will cluster around some curve called as the curve of regression. If the curve is a straight line it is called the line of regression and is called as linear regression, otherwise regression is said to be curvylinear.

★ Regression lines

If we take the case of two variables X and Y we shall have two regression lines, first regression of X on Y and second regression of Y on X.

The regression line of Y on X gives the most probable values of Y for given values of X and the Regression line X on Y is $X = a + bY$, (bxy) gives the most probable values of X for the given values of Y.

"The line of Regression is the line of best fit and is obtained by the principle of least square which consists of minimising the sum of the squares of deviations of the actual values of dependant variable from their estimated value."

Regression line of Y on X :-

Method 1 →

step 1: $Y = a + bX$ — ⊛

step 2: construct normal equations →

$$\Sigma Y = Na + b\Sigma X \quad \text{— ①}$$

$$\Sigma XY = a\Sigma X + b\Sigma X^2 \quad \text{— ②}$$

Step 3: calculate a and b from eqns ① and ② and put it in ⊛ gives the lines of regression of Y on X :-

$$Y = a + bX$$

Method 2) :- Regression line of Y on X is →

$$(Y-\bar{Y}) = b_{yx}(X-\bar{X})$$

where, $b_{yx} = \frac{\sigma_{xy}}{\sigma_x^2} = \frac{\sigma_x \sigma_y r}{\sigma_x^2}$

$$= r\frac{\sigma_y}{\sigma_x}$$

Regression line of x on y :-

Method I:

Step 1: $X = a + bY$ — (*)

Step 2: construct normal equation,

$$\Sigma X = Na + b\Sigma Y \quad — (1)$$

$$\Sigma XY = a\Sigma Y + b\Sigma Y^2 \quad — (2)$$

Step 3: Calculate a & b from (1) and (2) and put it in (*) gives the lines of ~~eqtn~~ regression of X on Y :-

$$X = a + bY$$

Method 2) :- Regression line of Y on X is →

$$(X-\bar{X}) = b_{xy}(Y-\bar{Y})$$

where, $b_{xy} = \frac{\sigma_{xy}}{\sigma_y^2} = \frac{r\sigma_x\sigma_y}{\sigma_y^2} = \frac{r\sigma_x}{\sigma_y}$

- Regressive coefficient

The slopes of the lines of best fit are called regression coefficient.

1) $b_{xy} = \frac{\sigma_{xy}}{\sigma_y^2} = \frac{r\sigma_x}{\sigma_y}$

2) $b_{yx} = \frac{\sigma_{xy}}{\sigma_x^2} = \frac{r\sigma_y}{\sigma_x}$

PAGE NO.:

- Properties of Regressive coefficient

1] Correlation coefficient is the geometric mean of regressive coefficient. $r = \pm\sqrt{b_{yx} \cdot b_{xy}}$

2] The correlation coefficient will have the same sign as that of regression coefficient.

3] Both the regression coefficients simultaneously cannot be greater than 1 i.e of one of the regression coefficient must be less than 1.

4] The modulus value of the arithmetic mean of the regression coefficient is not less than the modulus value of the correlation coefficient i.e $\left|\frac{1}{2}(b_{yx} + b_{xy})\right| > |r|$

5] Regression coefficients are independant of change of origin but not of scale.

Let $U = \frac{x-A}{h}$, $V = \frac{y-B}{k}$

$$b_{yx} = \frac{k}{h} b_{vu}$$

and $$b_{xy} = \frac{h}{k} b_{uv}$$

- Angle between two regression lines :-

$$\tan\theta = \frac{(1-r^2)}{|r|} \times \frac{\sigma_x \sigma_y}{(\sigma_x^2 + \sigma_y^2)}$$

Q

Q) Obtain the equation of two lines of regression for the following data. Also obtain the estimate of X for Y = 70

X	Y
65	67
66	68
67	65
67	68
68	62
69	62
70	69
72	71

$r = 0.271$

$\sigma_x = 2.1213$

$\sigma_y = 3.041$

$\sigma_{xy} = 1.75$

$\bar{x} = 68$

$\bar{y} = 66.5$

Method 2 →

Regression line of Y on X →

$$(Y-\bar{Y}) = b_{yx}(X-\bar{X})$$

$$b_{yx} = r\frac{\sigma_y}{\sigma_x} = \frac{0.271 \times 3.041}{2.1213} = 0.3725$$

$$(Y-66.5) = 0.372(X-68)$$

$$Y-66.5 = 0.372X - 25.33$$

$$Y = 0.372X + 41.165$$

Regression line of X on Y →

$$(X-\bar{X}) = b_{xy}(Y-\bar{Y})$$

$$b_{xy} = r\frac{\sigma_x}{\sigma_y} = \frac{0.271 \times 2.121}{3.041} = 0.189$$

$$X-68 = 0.189(Y-66.5)$$

$$X-68 = 0.189Y - 12.569$$

$$X = 0.189Y + 55.43$$

$\therefore$ If $Y = 70$

$x = 13.23 + 55.43$

$X = 68.66$

~~Method 1 →~~

$\cancel{\Sigma Y = Na + b\Sigma X}$

$\cancel{\Sigma XY = a\Sigma X + b\Sigma X^2}$

Method 1 →

Regression line of Y on X is →

$Y = a + bX$

Normal equation →

$\Sigma Y = Na + b\Sigma X$

$\Sigma XY = a\Sigma X + b\Sigma X^2$

eq ① → $532 = 8a + 544b$ –

eq ② → $36190 = 544a + 37028b$

~~[illegible]~~

eq ① × 68 → $36176 = 544a + 36992b$

$36190 = 544a + 37028b$

$-14 = 0 \cancel{+} 36b$

$b = 0.388$

$532 = 8a + 211.5$

$320.5 = 8a$

$a = 40.06$

Y on X → $Y = 40.06 + 0.38X$

Regression of X on Y →

$$X = a + bY$$

$$\Sigma X = na + b\Sigma Y$$

$$544 = 8a + 532b \quad — ①$$

$$\Sigma XY = a\Sigma Y + b\Sigma Y^2$$

$$36190 = 532a + 35452b \quad — ②$$

$a = 55.41$

$b = 0.189$

$$X = 55.41 + 0.189Y$$

The following table of 100 students gives according to age, the frequencies of marks

V		-2	-1	0	1	
U	Age y / Marks x	18	19	20	21	fx
-2	10-20	16 4	4 2	0 2	0 -	8
-1	20-30	10 5	4 4	0 6	-4 4	19
0	30-40	0 6	0 8	0 10	0 11	35
1	40-50	-8 4	-4 4	0 6	8 8	22
2	50-60	0 -	-2 2	0 4	8 4	10
3	60-70	0 -	-6 2	0 3	3 1	6
	fy	19	22	31	28	100

$\Sigma u_i f_i x_i' = 27$

X	fx	u	f_iU_i	u_i^2	$f_iu_i^2$
15	8	-2	-16	4	32
25	19	-1	-19	1	19
35	35	0	0	0	0
45	22	1	22	1	22
55	10	2	20	4	40
65	6	3	18	9	54
	100		25		167

y	fy	V	f_iV_i	v_i^2	$f_iv_i^2$
18	19	-2	-38	4	76
19	22	-1	-22	1	22
20	31	0	0	0	0
21	28	1	28	1	28
	100		-32		126

$A = 35$, $h = 10$

$B = 20$, $h = 1$

$$\therefore r_{uv} = \frac{\sigma_{uv}}{\sigma_u \sigma_v}$$

$$\sigma_{uv} = \frac{1}{N} \Sigma f_i u_i v_i - \bar{u}\bar{v}$$

$$\bar{u} = \frac{\Sigma f_i u_i}{\Sigma f_i} = \frac{25}{100} = 0.25$$

$$\bar{v} = \frac{\Sigma f_i v_i}{\Sigma f_i} = \frac{-32}{100} = -0.32$$

$$\sigma_{uv} = \frac{27}{100} + 0.08 = 0.35$$

$$\sigma_u = \frac{1}{N} \Sigma f_i u_i^2 - (\bar{u})^2 \qquad \sigma_v = \frac{1}{N} \Sigma f_i v_i^2 - (\bar{v})^2$$

$$= \frac{113}{100} - 0.0625 \qquad = \frac{126}{100} - 0.1024$$

$$= 1.67 - 0.0625 \qquad = 1.26 - 0.1024$$

$$= \cancel{1.19125} \ \cancel{1.0678} \qquad = \sqrt{1.1576} = 1.075$$

$$= \sqrt{1.6075} = 1.267$$

$$\therefore \cancel{r_{uv} = \frac{0.35}{1.0678 \times 1.1576}} = \frac{0.35}{1.236} = 0.283$$

$$\cancel{r_{uv} = \frac{0.35}{1.6075 \times 1.1576} = \frac{0.35}{1.86} = 0.188}$$

$$r_{uv} = \frac{0.35}{1.267 \times 1.075} = \frac{0.35}{1.362} = 0.2569$$

PAGE NO.:

Note: Both the Regression lines contain the point $(\bar{X}, \bar{Y})$ and intersect at $(\bar{X}, \bar{Y})$

Q1) In a partially destroyed lab. record of an analysis of correlation data the following results are available: variance of $X = 9$, Regression eq^{n}s are →

$$8X - 10Y + 66 = 0$$

$$40X - 18Y - 214 = 0$$

What are the values of :-

i) $\bar{X}, \bar{Y}$ ii) $r(X, Y)$ iii) σ_y iv) b_{yx} v) b_{xy}

2) Can $Y = 5 + 2.8X$ and $X = 3 - 0.5Y$ be the estimated regression eq^{n}s of Y on X and X on Y respectively?

3) For certain data of x and y which are correlated the two lines of $5x - 6y + 90 = 0$, $15x - 8y - 130 = 0$.

4) For 50 students of a class the regression eq^{n}s of marks of M1(X) on the marks of S and P (Y) is $3Y - 5X + 180 = 0$. The mean marks $\bar{Y} = 44$ and variance of X is $9/16^{th}$ of the variance of marks Y. Find $\bar{X} = ?$, $r(X, Y) = ?$

Q] Find rank correlation coefficient

MI	S&P	Rank X	Rank Y
X	Y		
9	8	1	1
6	5	4	3
7	3	3	5
8	2	2	6
5	1	5	7
4	4	6	4
3	6	7	2

Probability Distribution

- Probability distribution function :

- Q] Tossing of two coins simultaneously give following outcomes.

continuous RV

→ A

1st coin	2nd coin	
		HH
H	H	TT
T	T	HT
H	T	TH
T	H	

Sample space → S = {(H,H), (H,T), (T,T), (T,H)}

sample space → S = {HH, TH, HT, TT}

Q] Write the probability distribution of a random variable X: counts the number of heads occured in the tossing of two unbiased coins

⇒ Sample space : $\Omega = \{HH, TT, TH, HT\}$

$X : \Omega \to \mathbb{R}$

X: no. of heads occured in the experiment

$X(HH) = 2$

$X(TT) = 0$

$X(TH) = 1$

$X(HT) = 1$

Range set = {0, 1, 2}

$X = 0 : \{T, T\}$ ∴ $P(X=0) = 0.25$

$X = 1 : \{HT, TH\}$ ∴ $P(X=1) = 0.5$

$X = 2 : \{HH\}$ ∴ $P(X=2) = 0.25$

$X = x$	0	1	2
$P(X=x)$	0.25	0.5	0.25

event is subset of sample

Q] Find the probability distribution of the random variables.

X: sum of the two dice

Y: +ve difference of two dice

in tossing two dice.

→ Sample spaces : $\Omega = \{(1,1), (1,2), (1,3) \dots (6,6)\}$

$X=2 : \{1,1\}$

$X=3 : \{(1,2),(2,1)\}$

$X=4 : \{(1,3),(2,2),(3,1)\}$

$X=5 : \{(1,4),(2,3),(3,2),(4,1)\}$

$X=6 : \{(1,5),(2,4),(3,3),(4,2),(5,1)\}$

$X=7 : \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$

$X=8 : \{(2,6),(3,5),(4,4),(5,3),(6,2)\}$

$X=9 : \{(3,6),(4,5),(5,4),(6,3)\}$

$X=10 : \{(4,6),(5,5),(6,4)\}$

$X=11 : \{(5,6),(6,5)\}$

$X=12 : \{(6,6)\}$

$X=x_i$	2	3	4	5	6	7	8	9	10	11	12
$p(X=x_i)$	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{4}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$

$Y=0 : \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6)\}$

$Y=1 : \{(2,1),(3,2),(4,3),(5,4),(6,5)\}$

$Y=2 : \{(3,1),(4,2),(5,3),(6,4)\}$

$Y=3 : \{(4,1),(5,2),(6,3)\}$

$Y=4 : \{(5,1),(6,2)\}$

$Y=5 : \{(6,1)\}$

$Y=y_i$	0	1	2	3	4	5
$P(Y=y_i)$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$

* the function

Q] Find the probability distribution of no. of tails in tossing of three coins simultaneously

→ Sample space: $\Omega = \{HHH, TTT, HHT, HTH, THH, TTH, THT, HTT\}$

$X =$ no. of tail

$X = 0 : \{HHH\}$
$X = 1 : \{HHT, HTH, THH\}$
$X = 2 : \{TTH, THT, HTT\}$
$X = 3 : \{TTT\}$

$X = x_i$	0	1	2	3
$P(X = x_i)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

Q] A random variable X has the following probability function

$X = x$	0	1	2	3	4	5	6	7
$P(X = x)$	0	k	$2k$	$2k$	$3k$	k^2	$2k^2$	$7k^2+k$

i) Find k?
ii) Evaluate :- $P(X<6)$, $P(X \geq 6)$ & $P(0<X<5)$
iii) $P(X \leq a) \geq \frac{1}{2}$; Find the minimum value of a
iv) Determine cumulative distribution function of X.

i) $0 + k + 2k + 2k + 3k + k^2 + 2k^2 + 7k^2 + k = 1$

$9k + 10k^2 = 1$

$10k^2 + 9k - 1 = 0$

$10k^2 + 10k - k - 1 = 0$

$10k(k+1) - 1(k+1) = 0$

$k = -1$ or $\boxed{k = \frac{1}{10}}$

m_i or p_i pmf → probability mass function → probability distribution for discrete

$f(x)$ pdf → probability density function → continuous

Imp $F(x)$ CDF → cumulative $P(X \le x)$ — $\sum_{-\infty}^{x} p_i$ → discrete ; $\int_{-\infty}^{x} f(x)\,dx$ = continuous

PAGE NO.:

ii) $P(X<6) = P(X=0) + P(X=1) + \dots + P(X=5)$

$= 0 + k + 2k + 2k + 3k + k^2$

$= 8k + k^2$

$= \frac{8}{10} + \frac{1}{100}$

$\Rightarrow \frac{800+10}{1000} = \frac{810}{1000} = \frac{81}{100}$

$P(X \ge 6) = P(X=6) + P(X=7)$

$= 2k^2 + 7k^2 + k$

$= 9k^2 + k = \frac{9}{100} + \frac{1}{10} = \frac{90+100}{1000} = \frac{19}{100}$

$P(0<X<5) = P(X=1) + P(X=2) + P(X=3) + P(X=4)$

$= k + 2k + 2k + 3k$

$= 8k = \frac{8}{10}$

iv)

$X = x$	0	1	2	3	4	5	6	7
$P(X=x)$	0	1/10	2/10	2/10	3/10	1/100	2/100	17/100
CDF = $P(X \le x)$	0	$\frac{1}{10}$	$\frac{3}{10}$	$\frac{5}{10}$	$\frac{8}{10}$	$\frac{81}{100}$	$\frac{83}{100}$	1

$$F(X) = \begin{cases} 0, & X \le 0 \\ 1/10, & X \le 1 \\ 3/10, & X \le 2 \\ 1/2, & X \le 3 \\ 4/5, & X \le 4 \\ 81/100, & X \le 5 \\ 83/100, & X \le 6 \\ 1, & X \le 7 \end{cases}$$

iii) a ki aisi value jaha prob. $\frac{1}{2}$ is greater ho to $P(X \leq 3)$ me $a = 1/2$ hai isliye $a = 3$.

9] The diameter of an electric cable, say X is assumed to a continuous R.V with p.d.f $f(x) = 6x(1-x)$, $0 \leq x \leq 1$

1) Check that $f(x)$ is pdf and

2) determine a number k such that i.e. $P(X<k) = P(X>k)$

Q] If X be a continuous R.V with pdf given by

$$f(x)\begin{cases} kx & , \quad 0 \le x < 1 \\ k & , \quad 1 \le x < 2 \\ -kx + 3k & , \quad 2 \le x < 3 \\ 0 & , \quad \text{elsewhere} \end{cases}$$

i) Determine the constant k
ii) Determine the distribution function
iii) Find $P(0.5 < x \le 2.5)$

→ i) $\int_{-\infty}^{\infty} f(x)\,dx = 1$

$$\int_0^1 kx\,dx + \int_1^2 k\,dx + \int_2^3 (-kx) + (3k)\,dx = 1$$

$$\left[\frac{kx^2}{2}\right]_0^1 + [kx]_1^2 + \left[-\frac{kx^2}{2}\right]_2^3 + [3kx]_2^3 = 1$$

$$\frac{k}{2} + k + \left(-\frac{9k}{2} + \frac{4k}{2}\right) + (9k - 6k) = 1$$

$$\frac{k}{2} + k - \frac{5k}{2} + 3k = 1$$

$$-\frac{4k}{2} + 4k = 1$$

$$-2k + 4k = 1$$

$$k = \frac{1}{2}$$

iii) $P(0.5 < X < 2.5) = \int_{0.5}^{2.5} f(x)\,dx$

$$= \int_{0.5}^{1} kx\,dx + \int_1^2 k\,dx + \int_2^{2.5} (-kx + 3k)\,dx$$

$$= k\left[\frac{x^2}{2}\right]_{0.5}^{1} + k[x]_1^2 + -k\left[\frac{x^2}{2}\right]_2^{2.5} + 3k[x]_2^{2.5}$$

$$= k\left[\frac{1}{2}-\frac{1}{8}\right] + k - k\left[\frac{25}{4}-\frac{4}{2}\right] + 3k\left[\frac{5}{2}-2\right]$$

$$= k\left[\frac{6}{16}\right] + k - \frac{34k}{8} + \frac{3k}{2}$$

$$= \frac{6k}{10} + k + \frac{34k}{8} + \frac{3k}{2} = \frac{3k}{5} + k + \frac{17k}{4} + \frac{3k}{2} = \frac{147}{20}k$$

iii) C.D.F $\rightarrow$ $F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)\,dx$ ✓ ✗ ✗

$x \in (-\infty, 0) \Rightarrow F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)\,dx = 0$

$x \in (-\infty, 1) \Rightarrow F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)\,dx$

$$= \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{x} f(x)\,dx$$

$$= 0 + k\left[\frac{x^2}{2}\right]_0^x = \frac{x^2}{4}$$

Nai isme upper limit x rakhni hai !! to ye galat hai

sab ans same hai par writing galat hai!

$x \in (1, 2) \Rightarrow F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)\,dx$

$$= \int_{-\infty}^{0} f(x)\,dx + \int_{0}^{1} f(x)\,dx + \int_{1}^{x} f(x)\,dx$$

$$= \frac{k}{2} + kx - k = kx - \frac{k}{2} = \frac{x}{2}\ldots$$

$x \in (2, 3) \Rightarrow F(x) = P(X \le x) = \int_{-\infty}^{x} f(x)\,dx$

$$= \int_{-\infty}^{2} f(x)\,dx + \int_{2}^{3} f(x)\,dx$$

$$= \cancel{\frac{3}{4} + \frac{3}{2} - \frac{5}{4}}$$

$\cancel{= 0 + \frac{1}{2} + \frac{1}{2}}$

$$F(X) = P(X \le x) = 0 + \frac{1}{2} + \frac{1}{2} - k\left(\frac{x^2}{2} - \frac{4}{2}\right) + 3k(x-2)$$

$$= 1 - \frac{kx^2}{2} - 2k + 3kx - 6k + \cancel{\frac{4k}{2}}$$

$$= 1 - \frac{x^2}{4} - 1 + \frac{3x}{2} - 3 + \cancel{\frac{4k}{2}}$$

$$= -\frac{x^2}{4} + \frac{3x}{2} - \frac{5}{4}$$

$$x \in (3, \infty) \Rightarrow F(x) = P(X \le x) = \int_{-\infty}^{\infty} f(x)\,dx = 1$$

$\cancel{= 0 + \frac{1}{2}}$

$$F(x) = \begin{cases} 0, & -\infty \le x < 0 \\ x^2/4, & 0 \le x < 1 \\ 2x - 1/4, & 1 \le x < 2 \\ -x^2/4 + 3x/2 - 5/4, & 2 \le x < 3 \\ 1, & 3 \le x < \infty \end{cases}$$

Mean of random variable

→ • For discrete random variable:

$X = x \quad x_1 \quad x_2 \quad \ldots \ldots \quad x_n$

$P(X = x) \quad p_1 \quad p_2 \quad \ldots \ldots \quad p_n \longrightarrow$ pmf

mean of X is :- $\mu = \frac{\Sigma p_i x_i}{\Sigma p_i} = \Sigma p_i x_i$

• For continuous r.v:

~~xxx~~ $\quad X = x$

pdf $\quad f(x)$

mean of is : $\mu = \frac{\int_{-\infty}^{\infty} x\, f(x)\, dx}{\int_{-\infty}^{\infty} f(x)\, dx} = \int_{-\infty}^{\infty} x f(x)\, dx$

Variance of r.v

For discrete r.v :- $\quad X \quad x_i$

$P(X = x) \quad p_i$

$$\sigma^2 = \Sigma p_i (x_i - \mu)^2$$

continuous r.v → $\quad X = x$

pdf → $f(x)$

$$\sigma^2 = \int_{-\infty}^{\infty} f(x)(x - \mu)^2\, dx$$

prob. jaha $\frac{1}{2}$ aati hai wo median

jiski prob. jyada wo mode

PAGE NO.:

Q] Let X be a r.v giving the no. of heads in three toses of fair coins
Find → 1) Mean (X) 2) Var (X) 3) σ_x

→ Probability distribution is given by:

$X=x$	0	1	2	3
$P(X=x)$	$\frac{1}{8}$	$\frac{3}{8}$	$\frac{3}{8}$	$\frac{1}{8}$

$$\mu_x = \Sigma p_i x_i = 0\times\frac{1}{8} + 1\times\frac{3}{8} + 2\times\frac{3}{8} + 3\times\frac{1}{8} = \frac{3+6+3}{8} = \frac{12}{8} = \frac{3}{2}$$

$$\sigma_x^2 = \Sigma p_i (x_i - \mu)^2$$

$$= \cancel{0\times\frac{-11}{8} + 1\times\left(\frac{-9}{8}\right) + 2\times\left(\frac{-9}{8}\right) + 3\times\left(\frac{-4}{8}\right)}$$

$$= \cancel{\frac{-21}{8}}$$

$$= \frac{1}{8}(0-1.5)^2 + \frac{3}{8}(1-1.5)^2 + \frac{3}{8}(2-1.5)^2 + \frac{1}{8}(3-1.5)^2$$

$$= 0.28125 + 0.09375 + 0.9375 + 0.28125$$

$$= \frac{3}{4}$$

$$\sigma_x = \sqrt{\sigma_x^2} = \frac{\sqrt{3}}{2}$$

Q] $f(x) = \begin{cases} e^{-x} & , x \geq 0 \\ 0 & x < 0 \end{cases}$

i) μ_x ii) σ_x^2 iii) σ_x

$$\mu = \int_{-\infty}^{\infty} x f(x)\,dx = \int_{-\infty}^{0} (x\times 0)\,dx + \int_{0}^{\infty} x e^{-x}\,dx$$

$$= 0 + -\left[x e^{-x}\right]_0^\infty + \int_0^\infty e^{-x}\,dx$$

$$\mu = -\left[\lim_{x\to\infty} \frac{x}{e^x} - 0\right] + \int_0^\infty e^{-x}\,dx$$

$$= -\left[\lim_{x\to\infty} \frac{1}{e^x} - 0\right] + -[e^{-x}]_0^\infty$$

$$= 0 - \left[\lim_{x\to\infty} e^{-x}\right] + e^{-0}$$

$$= -\lim_{x\to\infty} \frac{1}{e^x} + 1$$

$$= 0 + 1 = 1$$

$$\sigma_x^2 = \int_{-\infty}^{\infty} (x-1)^2 f(x)\,dx$$

$$= \int_{-\infty}^{0} (x-1)^2\, 0\,dx + \int_0^\infty (x-1)^2 f(x)\,dx$$

$$= \int_{-\infty}^{0} 0\,dx + \int_0^\infty (x-1)^2 e^{-x}\,dx$$

$$= \int_0^\infty (x-1)^2 e^{-x}\,dx$$

$$\sigma_x^2 = \left[-(x-1)^2 e^{-x}\right]_0^\infty + \int_0^\infty 2(x-1)e^{-x}\,dx$$

$$= -\left[\lim_{x\to\infty} \frac{(x-1)^2}{e^x}\right] + \left[-2(x-1)e^{-x}\right]_0^\infty + 2\int_0^\infty e^{-x}\,dx$$

$$= \lim_{x\to\infty} \frac{2(x-1)}{e^x} - \left[\lim_{x\to\infty} \frac{2(x-1)}{e^x}\right] + 2\left[-e^{-x}\right]_0^\infty + 1$$

$$= 2\lim_{x\to\infty} \frac{1}{e^x} + 1$$

$$= \cancel{2}\ 1$$

$$\sigma_x = \sqrt{1} = 1$$

PAGE NO.:

Q] Suppose two dice are tossed, find:
1) mean 2) variance of a r.v X denotes the sum of the two no.'s on two dice

→

$X = x_i$	2	3	4	5	6	7	8	9	10	11	12
p_i	$\frac{1}{36}$	$\frac{1}{18}$	$\frac{1}{12}$	$\frac{1}{9}$	$\frac{5}{36}$	$\frac{1}{6}$	$\frac{5}{36}$	$\frac{1}{9}$	$\frac{1}{12}$	$\frac{1}{18}$	$\frac{1}{36}$

3) Find first four moments about origin
4) ———— 11 ———— about mean

Q) Find the probability distribution of no. of tails in tossing of ~~10~~ 5 unfair coins, where the probability of getting tail in each coin is 0.7.

⇒

BINOMIAL DISTRIBUTION / BERNOULLY

$P(X=x) = {}^nC_x\, p^x q^{n-x}$

$\Omega = \{S, F\}$

$P(S) = p$ and $P(F) = q$

also $p+q=1$

$\therefore q = 1-p$

⇒ $X=x$	0	1	2	3	4	5
$P(X=x)$	0.00243	0.02835	0.1323			

$0 \rightarrow {}^5C_0 (0.7)^0 (0.3)^5 = 0.00243$

$1 \rightarrow {}^5C_1 (0.7)^1 (0.3)^4 = 0.02835$

$2 \rightarrow {}^5C_2 (0.7)^2 (0.3)^3 = 10 \times 0.49 \times 0.027 = 0.1323$

$3 \rightarrow {}^5C_3 (0.7)^3 (0.3)^2 = 10 \times 0.343 \times 0.09 = 0.3087$

$4 \rightarrow {}^5C_4 (0.7)^4 (0.3)^1 = 0.36015$

$5 \rightarrow {}^5C_5 (0.7)^5 (0.3)^0 = 0.16807$

PAGE NO.:

consider a random experiment with only two possible outcomes $\Omega = \{S, F\}$, let p denote the prob. of success, q = ~~denote~~ 1-p denotes prob. of failure. Assume that exp. is independantly repeated ~~and~~ n times and X denotes a random variable which counts the ~~n~~ no. of successes in n trials. Thus the value of X can have 0, 1, 2, 3 .... m then prob dist. of this random variable X is given by the prob. function,

$$P(X=x) = {}^{n}C_{x}\, p^{x} q^{n-x}$$
$$x = 0, 1, 2, 3 \ldots n$$

Properties:

If X is binomially distributed then,

1) mean(μ) of the dist. $X = np$
2) variance of X is $\sigma^2 = npq$
2) coefficient of skewness $= \dfrac{q-p}{\sqrt{npq}}$
4) coefficient of kurtosis $= 3 + \dfrac{1-6pq}{\sqrt{npq}}$

For binomial mean $= np = \sum_{x=0}^{n} x \cdot {}^nC_x p^x q^{n-x}$

variance $= npq = \sum {}^nC_x p^x q^{n-x} (x_i - \mu)^2$

PAGE NO.:

Exponential Distribution

A continuous random variable x is said to be exponentially distributed if its pdf is given by:

$$f(x) = \begin{cases} \alpha e^{-\alpha x}, & x > 0 \\ 0, & x \leq 0 \end{cases}$$

Properties:

mean: $\mu = \frac{1}{\alpha} = \int_{-\infty}^{\infty} x f(x) dx = \int_{0}^{\infty} \alpha e^{-\alpha x} dx$

variance: $\sigma^2 = \frac{1}{\alpha^2}$

Uniform Distribution

A continuous random variable x is said to be uniformly distributed in $x \in [a, b]$ if its pdf is given by:

$$f(x) \begin{cases} \frac{1}{b-a}, & a \leq x \leq b \\ 0, & \text{else} \end{cases}$$

Properties: $\mu = \frac{a+b}{2}$

$$\sigma^2 = \frac{(b-a)^2}{12}$$

Normal Distribution

A cont. r.v. X is said to be normally distributed if its pdf is given by:

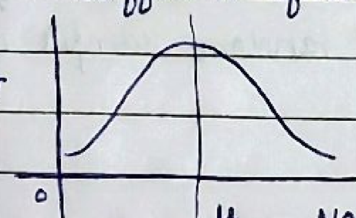
cumulative frequency mela ogive

$$f(x) = \frac{1}{\sqrt{2\pi} \times \sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}, -\infty < x < \infty$$

where μ = mean, σ = standard deviation

Props → μ = mean
variance $= \sigma^2$
coefficient of skewness = 0
coefficient of kurtosis = 3

X →



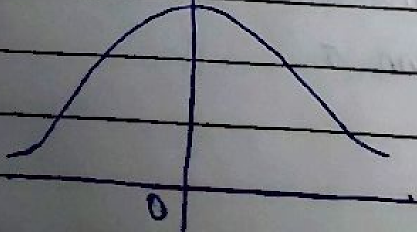
Normal curve

- Standardized variable (z) for normally distributed r.v X :-

$$Z = \frac{X - \mu}{\sigma}$$

mean = 0 and variance = 1
Z is called standardized normal r.v with mean = 0 and variance = 1, having pdf :-

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}$$



Q1] If ~~a~~ the diameters of the ball bearings are normally distributed with mean = 15.6 mm and sd = 0.06 mm determine the percentage of ball bearings with diameters,

* 1) between 15.50 to 15.70 mm

2) greater than 15.70 mm

3) less than 15.40 mm

4) equal to 15.60 mm

Assume the measurements to be recorded to the nearest 0.01 mm

Q2] A machine produces bolts which are 10% defective, find the probability that in a random sample of 400 bolts produced by the machine,

* 1) between 30 and 50

2) atmost 30

3) 55 or more will be defective

Relation between Binomial and Normal distribution

If the number of trials n in the binomial distribution is large and the prob. of success and failure that is p and q are not too close to 0 then binomial dist. can be closely approximated by normal distri. with mean $\mu = np$ and $\sigma = \sqrt{npq}$

NOTE: For better approximations we must have np and nq both greater than 5.

Poisson distribution

If a discrete r.v x has the following probability distribution

$$P(X=x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2 \ldots$$

where λ is given positive constant, and x is called as poisson distribution.

Prop. :→ ① The mean $\mu = \lambda$

② The variance $\sigma^2 = \lambda$

③ Coefficient of skewness $= \frac{1}{\sqrt{\lambda}}$

④ Coefficient of kurtosis $= 3 + \frac{1}{\lambda}$

1 Relation between Binomial & Poisson distribution,

→ If the number of n trials in the binomial distribution is large and the probability of occurence of s i.e $P(S) = p$ is closer to 0 i.e q is closer to 1, then the binomial distribution can be closely approximated by the poisson distribution with mean, $\mu = np = \lambda$

Recap→Q] In tossing of 10 dice, if a r.v x counts the no. of even no. appearance on these dice. Find the probability distribution:

1) Find mean

2) Find variance

3) Find coefficient of skewness

4) Find coefficient of kurtosis

5) Find probability of getting atmost two even no. on dice